Introduction to CSS (Cascading Style Sheets)

What is CSS?

CSS (Cascading Style Sheets) is a stylesheet language used to describe the **presentation and layout** of HTML documents.

If:

- **HTML = Structure (Skeleton)**
- **CSS = Design (Skin + Appearance)**
- **JavaScript = Behavior (Brain)**

Then CSS controls:

- Colors
- Fonts
- Spacing
- Layout
- Positioning
- Responsive design
- Animations

Why Do We Need CSS?

Without CSS, web pages look plain and unstructured.

- Without CSS

- With CSS

👉 Same content — different presentation.
That transformation is done using CSS.

What Does “Cascading” Mean?

The word **Cascading** means:

Styles can come from multiple sources and follow a priority system.

CSS follows this priority (highest to lowest):

1. Inline CSS
2. Internal CSS
3. External CSS
4. Browser default styles

This priority system is called the **Cascade Rule**.

Ways to Apply CSS

Inline CSS

- Applied directly inside HTML tag.

```
<p style="color:red;">Hello World</p>
```

✓ Quick testing
❌ Not recommended for large projects

Internal CSS

- Defined inside <style> tag in <head> section.

```
<head>
  <style>
    p {
      color: green;
    }
  </style>
</head>
```

✓ Good for small websites
❌ Not scalable

External CSS (Recommended)

- CSS is written in a separate .css file.

```
<link rel="stylesheet" href="styles.css">
```

styles.css

```
p {
   color: purple;
}
```

✓ Clean
✓ Reusable
✓ Industry standard
✓ Maintainable

CSS Selectors (Very Important Concept)

- Selectors define which HTML elements to style.

Common Selectors:

Selector	Example	Meaning
Element	p	All paragraph tags
ID	#header	Specific element with id
Class	.box	All elements with class
Universal	*	All elements

Basic Syntax of Selector

```css
selector {
    property: value;
}
```

Example:

```css
p {
    color: blue;
    font-size: 18px;
}
```

```css
#main {
    background-color: yellow;
}

.box {
    border: 2px solid black;
}
```

Complete CSS Selectors

Selector	Example	Description
Universal	* {}	Selects all elements
Element	p {}	Selects all <p>
Class	.box {}	Selects elements with class

Selector	Example	Description
ID	#header {}	Selects unique ID
Group	h1, h2 {}	Multiple selectors
Descendant	div p {}	Inside element
Child	div > p {}	Direct child
Attribute	input[type="text"]	Based on attribute
Pseudo-class	a:hover	State-based
Pseudo-element	p::first-letter	Part of element

CSS Versions

CSS has evolved over time:

- **CSS1 (1996)**
- **CSS2 (1998)**
- **CSS3 (Modular approach)**

Modern web development heavily uses CSS3 features like:

- **Flexbox**
- **Grid**
- **Animations**
- **Media Queries**
- **Transforms**